



Break In Procedure

1. BREAK IN PROCEDURE: To avoid premature wear of gears and bearings, all new and rebuilt transmissions must be properly run in prior to racing. Drive slowly in the pits loading and unloading the transmission in each gear to lap in the gears and lubricate the bearings thoroughly. After accomplishing the above, drain the synthetic break in oil and replace with new synthetic gear oil as recommended for your type of transmission, Synthetic Multi-Viscosity Gear Oil 75W90. It is recommended that after the first event you drain the gear oil to remove all break-in debris and metallic sediment. Replace with new oil as recommended; thereafter, drain and inspect the oil and gears as often as you would inspect your engine, and whenever it produces a foul smell from heat.

2. GEAR OIL: SYNTHETIC MULTI-VISCOSITY GEAR OIL MUST BE USED
OVAL/ ROAD RACE: 75W90
DRAG RACE: 75W90

Do not at any time use straight 90W gear oil.

DO NOT OVERFILL! This will cause excessive heat.

Cases (Top Loader): Fill the transmission approximately 3/4" bellow the side fill hole.

Cases (Top and Bottom Loader): Fill the transmission approximately 3/4" bellow the side fill hole.

The split case 5-speed, WC4-4, WC4-4.2 requires one (1) quart, all other style of transmissions require approximately two (2) quarts.

3. PRE RACE: Treat your transmission the same as your engine. It is recommended that the transmission and the complete drive train have heat in them prior to racing. To do so, lift the rear wheels above the ground by placing a jack under the rear end, allowing the drive train to operate in high gear. Remember; never allow the drive train to operate with the car supported on jack stands and the rear hanging below ride height, as this will accelerate U-joint wear.

4. DETERMINING WHETHER OR NOT YOU NEED A COOLER OR FILTER:

Oval road race transmissions only: On a road course, a cooler is a must.
For the short-track and super speedways: Depending on the design and/or build of your car, and the airflow to the transmission, you may not need a cooler.

2 Speed, 3 Speed, and Drag Race (4 and 5 Speed) do not need a cooler.

5 Speed Road Race does require a cooler.

5. INSTALLATION / MOUNTING: If a rear cross member mount is used it must be a flexible rubber mount. Do not use a solid transmission mount. If the motor is mounted both front and rear using engine plates, then no rear transmission mount is required; but if you use one, make sure that it is a very soft mount. The transmission mount should not keep the engine from moving front to rear. Add lateral bars to the engine to keep it from moving.

6. BELLHOUSING ALIGNMENT: Bell housing alignment and pilot bearing hole depth must be confirmed before installation; failure to do so will result in premature wear of bearings, gears and clutch.

FOR CLUTCHLESS DRAG RACE TRANSMISSIONS ONLY

IF YOU HAVE PURCHASED ONE OF OUR
DRAG RACE CLUTCHLESS TRANSMISSIONS,
WE HAVE FURTHER INSTRUCTIONS AND WARNINGS AS FOLLOWS:

Although our drag race clutch less 4 speed offers the maximum in 4 speed performance, there are many things you must first understand in achieving the most from this gear box:

With the many clutch pressure plate and clutch disc combinations available today, it is impossible for us to recommend one particular type or make over another, just as it is impossible for us to dictate methods of dialing in a clutch combination to achieve the most in your particular installation and use.

YOU MUST ABORT PASSES WHEN YOUR CLUTCH DOES NOT HAVE ENOUGH SLIPPAGE.

For the gearbox to survive,

CLUTCH SLIPPAGE IS A MUST.

THE CLUTCH MUST SLIP DURING GEAR CHANGES.

In addition, it is a very fine line between too much slippage and not enough. While many clutch manufacturers recommend starting at the highest clutch spring pressures and backing off until minimal holding pressure is obtained, we recommend starting at lower spring pressures and dialing in more spring pressure until minimal holding pressure is obtained. This is necessary to prevent excessive shock to the gear systems within the transmission.

Dialing in the clutch is a delicate, time-consuming process that requires considerable thought. If necessary, you should start with only first and second gear burnouts prior to any full-power complete passes.

THERE IS NO WARRANTY STATED OR IMPLIED DUE TO THE UNUSUAL STRESSES PLACED ON RACING PARTS AND BECAUSE WE HAVE NO CONTROL OVER HOW THEY ARE USED OR INSTALLED.